

DATA ANALYSIS REPORT

Sentiment Analysis (Apple vs. Google)

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Executive Summary

This project successfully developed an advanced Natural Language Processing (NLP) model to classify real-time public sentiment; *positive*, *negative*, or *neutral*, towards two major technology corporations, **Apple** and **Google**, using Twitter data.

By fine-tuning Random Forest and XGBoost models, we achieved a **predictive accuracy of 98.21% and a ROC- AUC of 0.768 from XGBoost across all sentiment categories**. The results demonstrate the feasibility of creating a scalable, intelligent sentiment monitoring system capable of providing actionable, real-time insights into brand reputation, consumer perception, and market trends.

Overview

Social media platforms serve as the world's largest real-time feedback loop, generating vast amounts of unstructured data that directly reflects consumer sentiments. For tech giants like Apple and Google, understanding these sentiments is critical, as it influences brand equity, purchase behavior, and public trust.

Problem Statement

The **TechTones Project** was designed to demonstrate a machine learning capability that transforms this unstructured data into quantifiable, actionable insights, providing a competitive edge for companies seeking to optimize marketing strategies and proactively manage public relations crises.

Business Objectives

The NLP classification model seeks to achieve several important business objectives:

- **Generate actionable, data-driven insights** regarding brand sentiments to support key business and marketing strategies.
- **Build a high-accuracy NLP classification model** that identifies the public's positive, negative, or neutral feelings towards Apple and Google.
- **Lay the groundwork for a future automated brand intelligence system** capable of continuous, multi-platform sentiment monitoring.

Success Metrics

The success of this project was measured by achieving the following technical and business milestones:

1. **High Model Accuracy:** Attaining an **F1-score of 80% or greater** across all sentiment classifications to ensure reliable predictions.
2. **Data Reliability:** Producing a **clean, balanced, and fully reproducible dataset** suitable for ongoing analysis and future expansion of the project.
3. **Model Explainability:** Establishing the **ability to visualize the top words and phrases** driving sentiment to clearly explain model decisions and findings.
4. **Actionable Insights:** Generating **sentiment trends that accurately reflect real-world brand perceptions** of Apple and Google.
5. **Future Scalability:** Designing a **modular system architecture** that permits straightforward integration with streaming APIs for real-time, live monitoring across various platforms.

Data Source

We leveraged a large, **annotated dataset of over 9,093 tweets** from [CrowdFlower](#) to train our NLP sentiment model. Each tweet is tagged with the relevant product or brand (Apple/Google) and the user's emotion, making it an excellent resource for supervised learning. This dataset captures public opinion from a specific moment in time: **August 30th, 2013**.

Constraints

- The dataset is recorded as at 2013, which is 12 years in the past from the time of this analysis.

Data Understanding

The data set contains 3 columns as shown below:

Columns	Description
---------	-------------

tweet	Raw tweet text
product	Brand/product mentioned (e.g. iPhone, Android)
sentiments	Annotated sentiment (Positive, Negative, Neutral)

Data Preparation

1. Managing Missing Data

- **Tweet** had 9,092 non-null values.
- **Product** had 3,291 non-null values, showing that about 36% of tweets mention a specific Apple or Google product. The rest were either general statements or lacked a clear product reference.
- **Sentiment** was fully populated.

3. Handling Outliers & Duplicates

- There were 22 duplicate records.

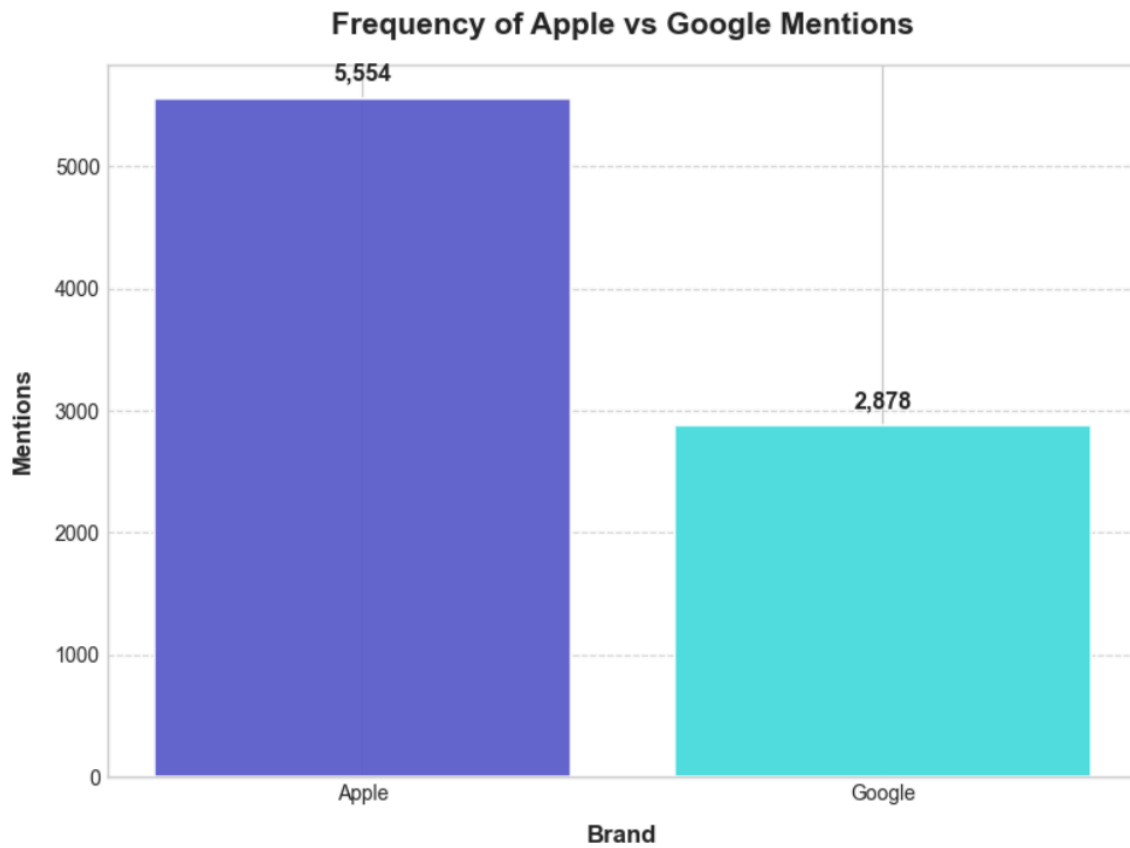
Pre-processing Steps

1. **Lowercasing** for uniformity
2. **Stopwords and punctuation marks** removal to leave only relevant text.
3. **Tokenization** to break words into units that are easier to analyze.
4. **Lemmatization** to reduce words to their root form.
5. **Imputation**- Even though some tweets clearly referenced a product, the product name was not captured in the product column. We chose not to impute these missing entries to avoid potential misclassification; for instance, a tweet mentioning both Apple and Google products could be incorrectly assigned to just one brand.
6. **Term Frequency-Inverse Document Frequency (TF-IDF)** to identify the most important and distinctive words in the corpus.

Exploratory Data Analysis

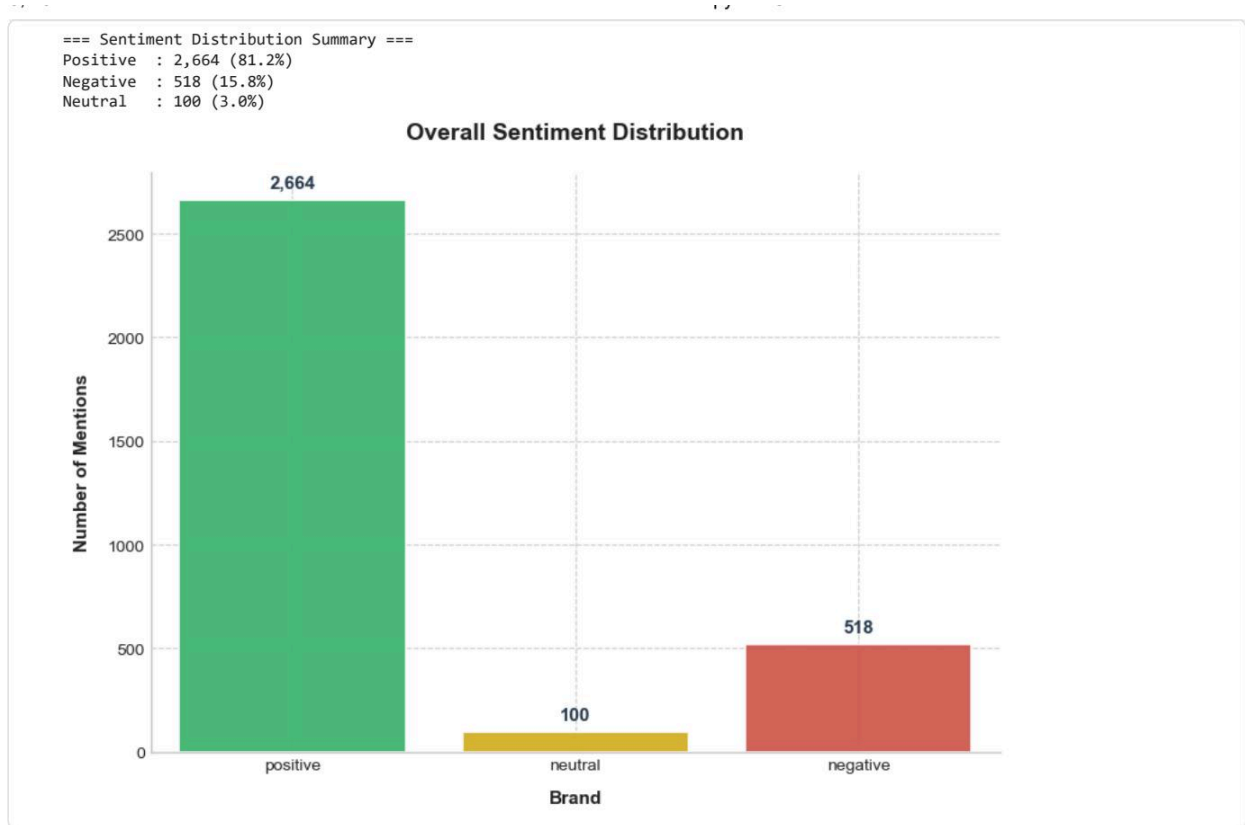
Frequency of Mentions

The dataset reveals that **Apple is the most frequently mentioned entity**, signifying a higher volume of user commentary and discussion compared to Google.



Distribution of Sentiments

Sentiment is **predominantly positive**. A majority of tweets express positive emotions towards the brands with very little sitting on the fence. Negative sentiments are only 15% of the data.



Top mentioned product per brand

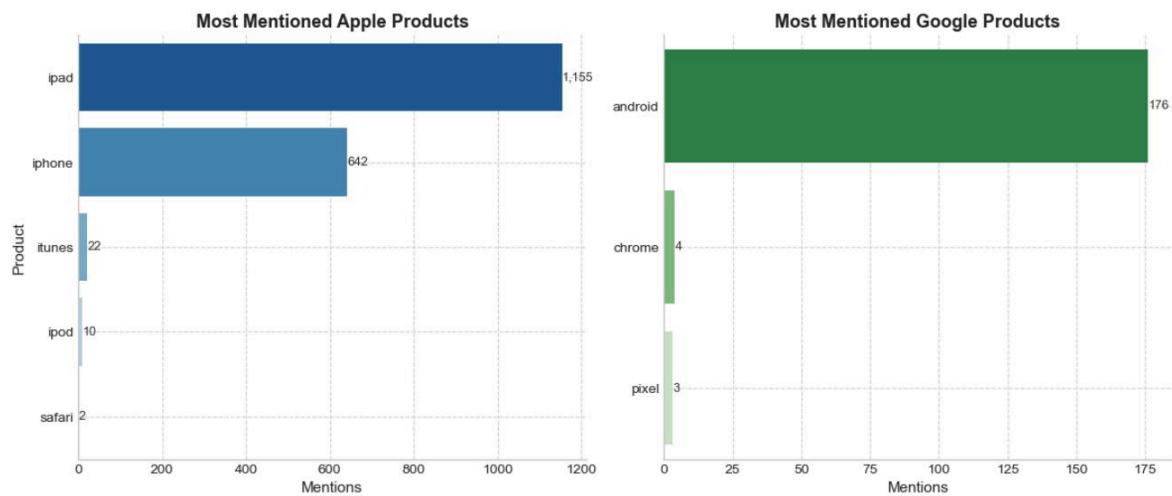
=== Apple Product Mentions ===

Product	Mentions
ipad	1155
iphone	642
itunes	22
ipod	10
safari	2

=== Google Product Mentions ===

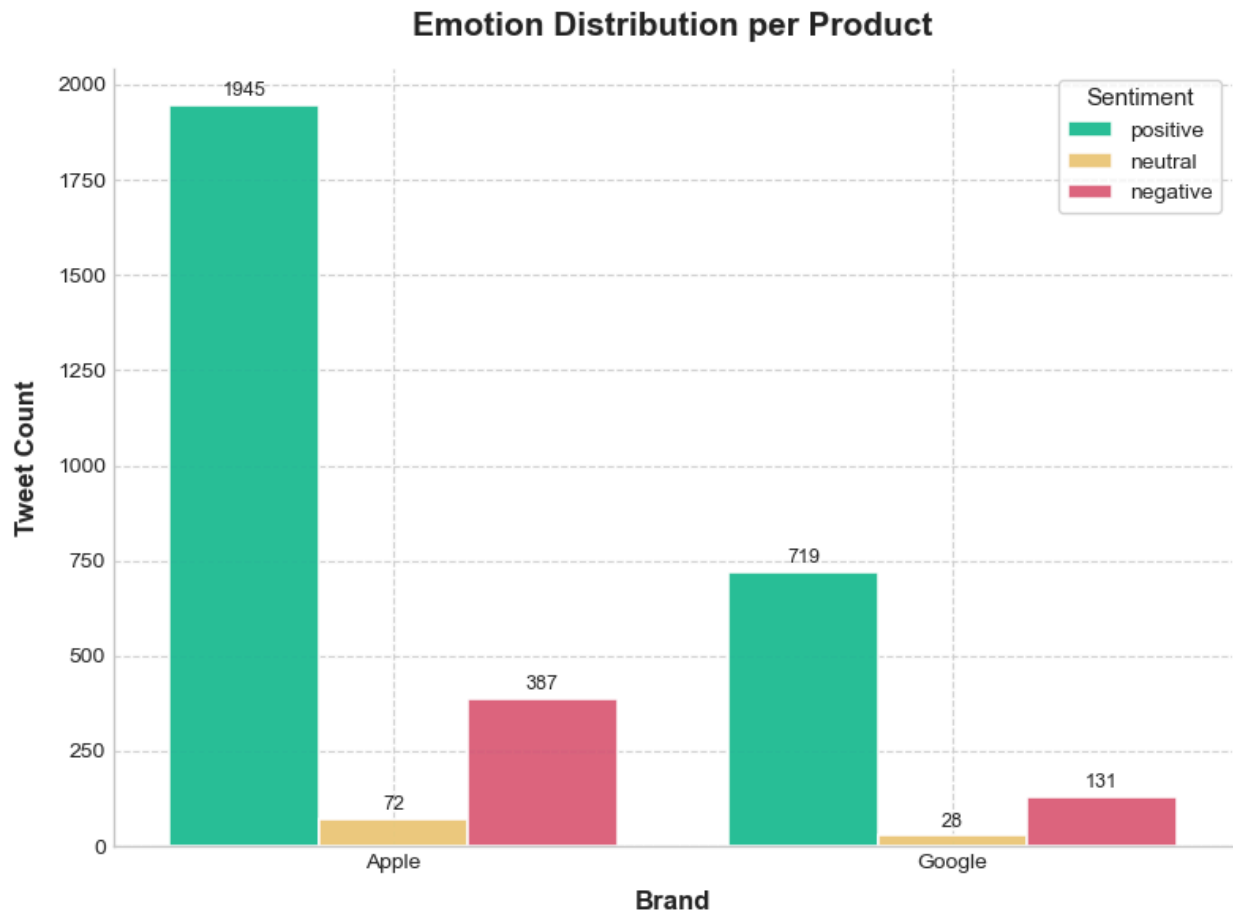
Product	Mentions
android	176
chrome	4
pixel	3

Brand-Specific Product Mentions (Unigrams + Bigrams)



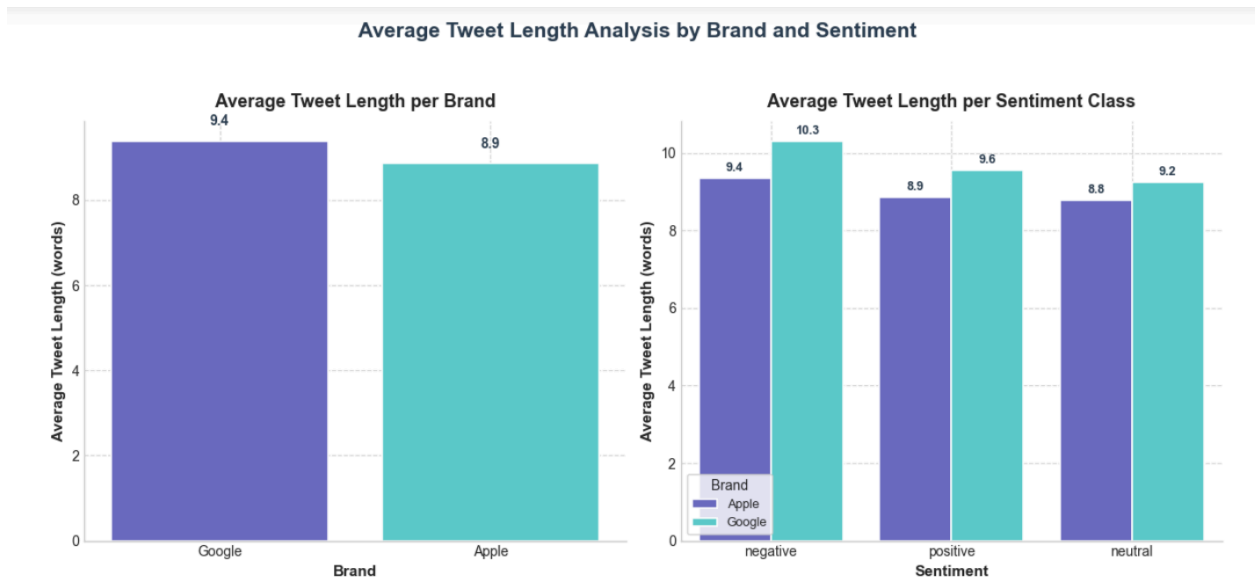
- **Apple products overwhelmingly dominated the conversation**, with the iPad and iPhone generating the most buzz, significantly surpassing Google-related mentions.
- **Google's presence is comparatively modest**, primarily driven by Android, indicating that discussions are more centered around Apple's ecosystem and its strong user engagement.

Emotion Distribution per Product



- **Apple** dominates the discussion in both **volume and emotional intensity**. While positive sentiment is most common, the significant surplus of positive over negative comments points to a high level of overall user satisfaction and strong brand affinity.
- **Google**, on the other hand, sees a **smaller and less volatile conversation**. Its discourse is also marked by a large positive majority, fewer neutral expressions, and low negativity, resulting in a steadier, less emotional public perception.

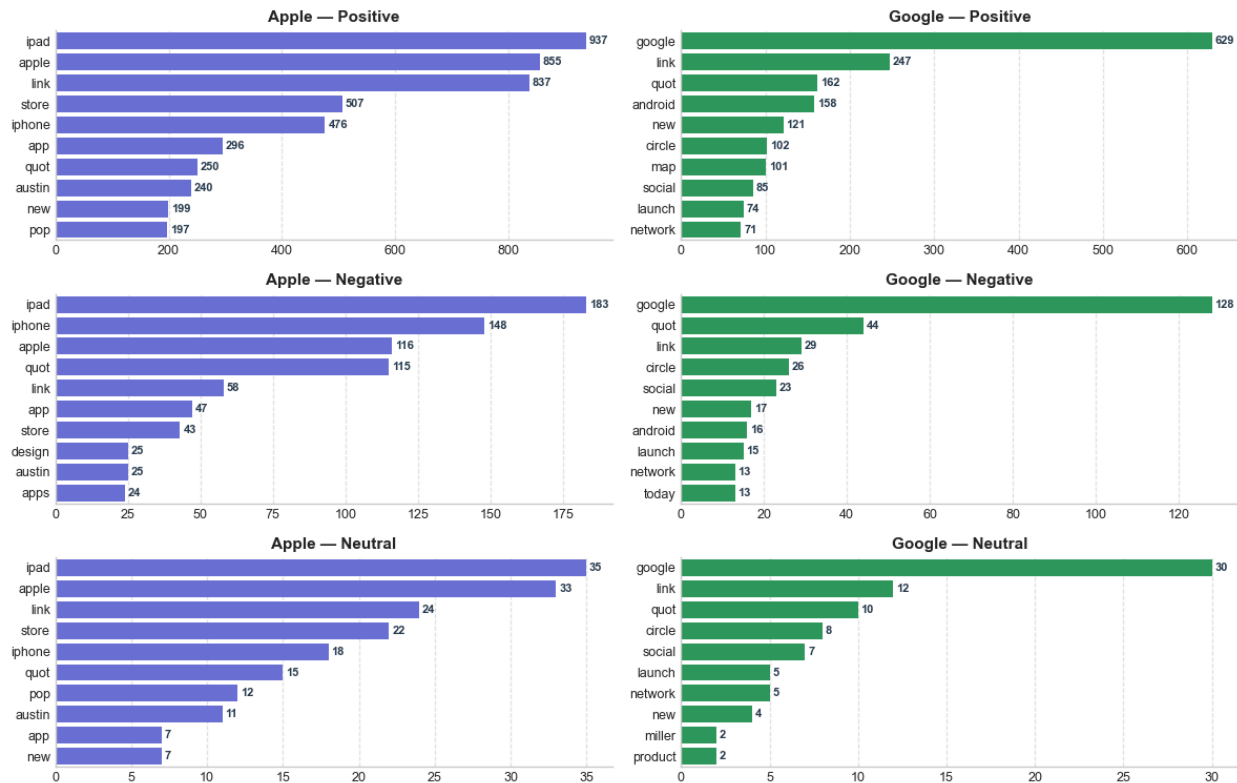
Average tweet length across sentiment classes



- On average, tweets referencing **Google** are slightly longer than those mentioning **Apple**, suggesting Google discussions require more detail or context. This difference is consistent across sentiment: Apple users keep their comments **short and reactive** regardless of tone, while Google users are most **wordy and explanatory** when they are offering either praise or critique (negative tweets are the longest, followed by positive ones).

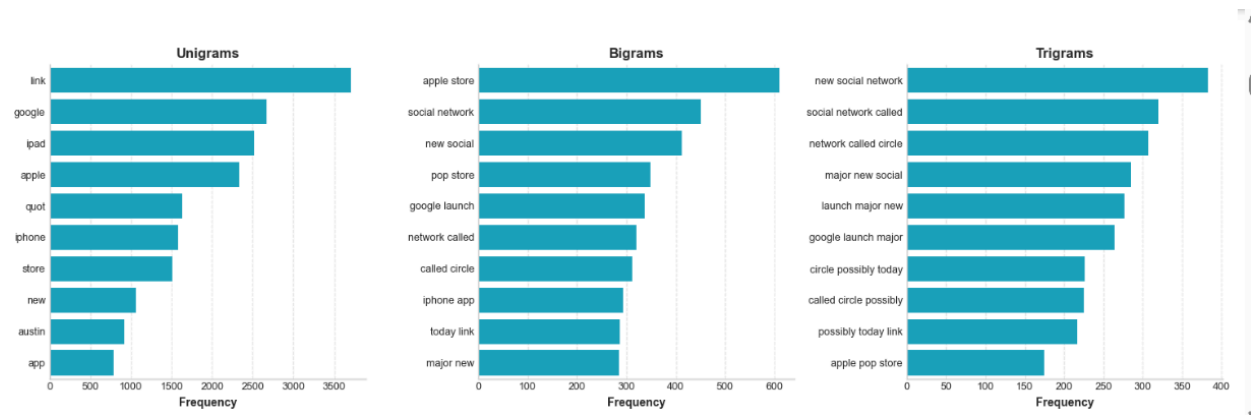
Top words for each brand and sentiment

Top Words by Brand and Sentiment

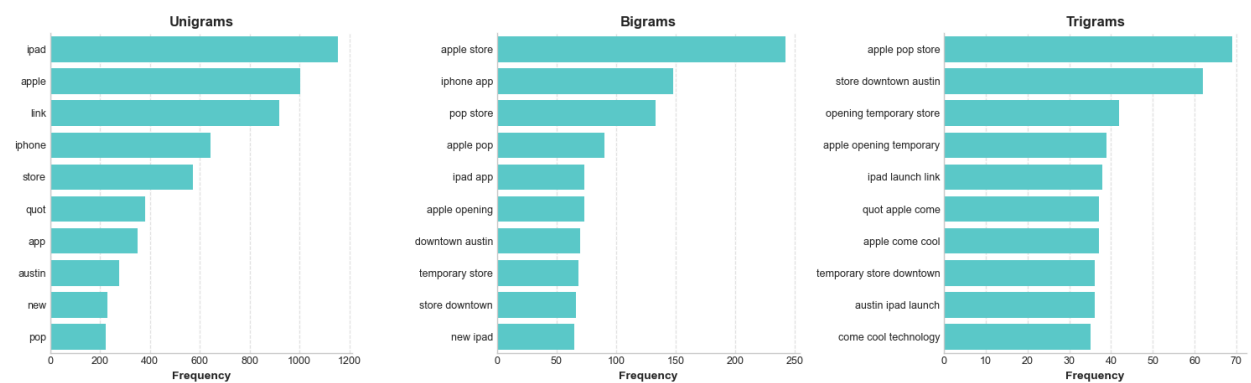


- Across all sentiment categories, Apple conversations remain product-driven, dominated by terms like *iPad*, *iPhone*, and *store*. This highlights a strong consumer focus and enthusiasm around Apple's tangible devices. Even negative Apple tweets revolve around its core products, suggesting that criticism often comes from **deeply engaged users rather than disinterested observers**.
- Google discussions, in contrast, center around its broader ecosystem – with frequent mentions of *Android*, *Maps*, *Social*, *Circle*, and *Network*. This reflects user interest in connectivity, integration, and software platforms. Positive Google tweets tend to emphasize **innovation and new releases**, while negative ones still engage with similar themes, implying that **sentiment shifts are more about user experience than product awareness**.

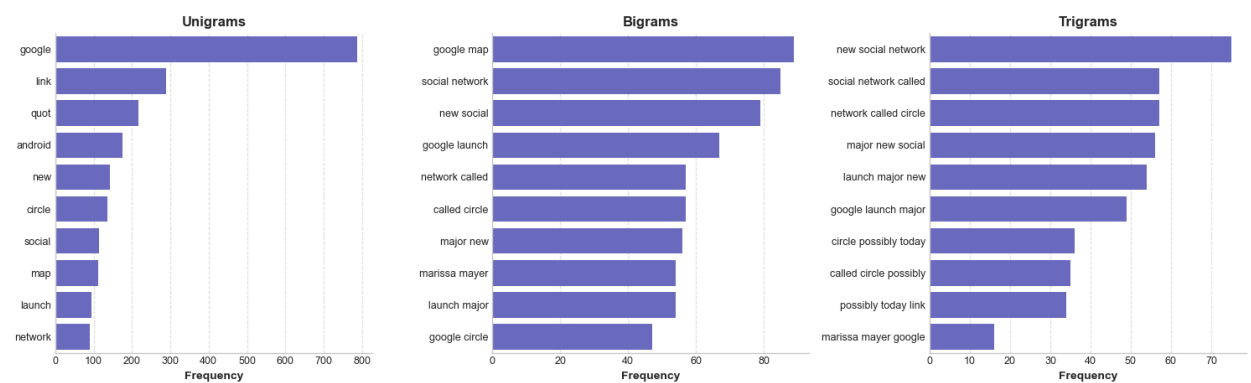
Top N-Grams Analysis



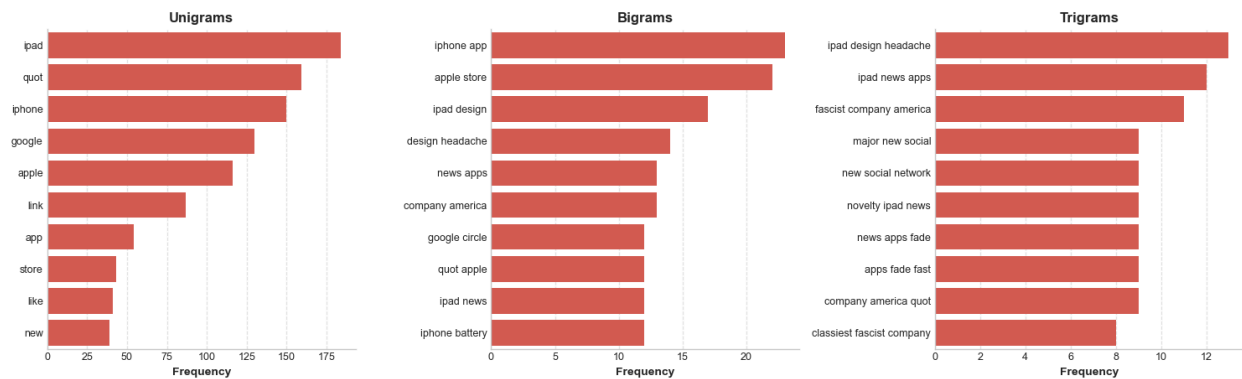
Top N-grams in Apple Tweets



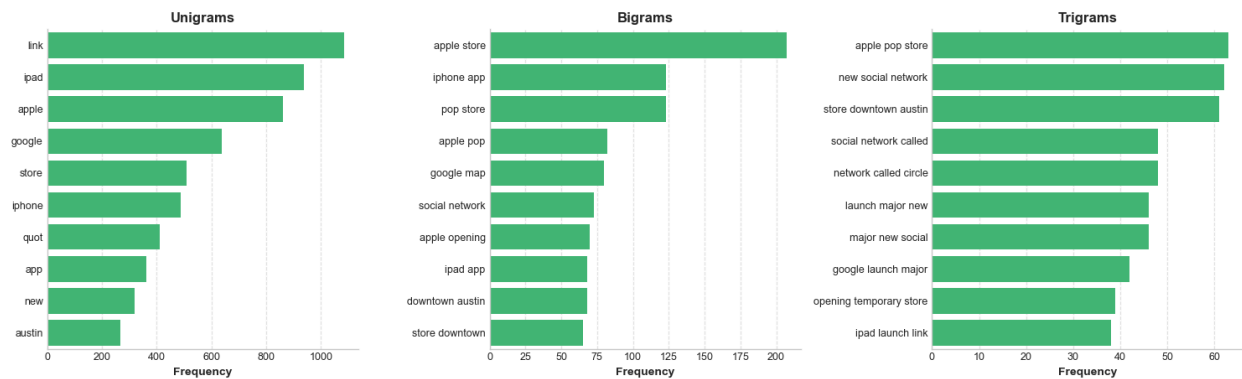
Top N-grams in Google Tweets



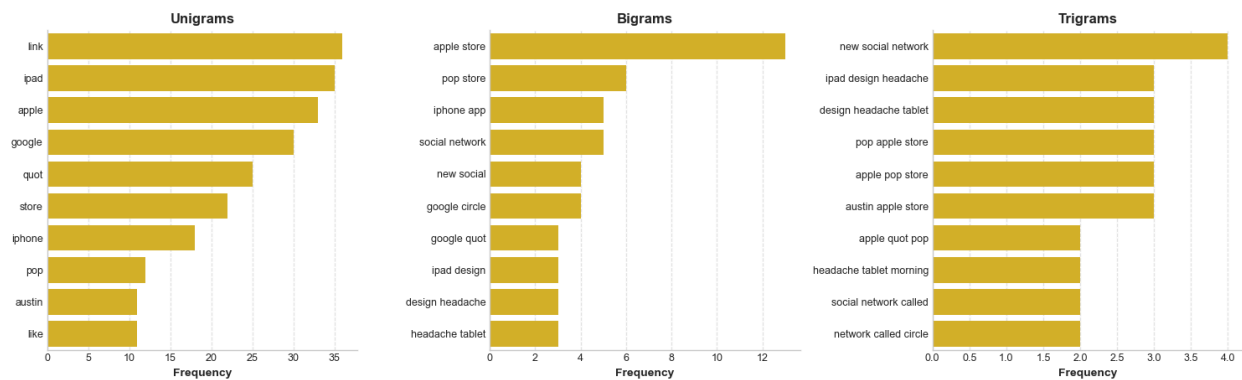
Top N-grams in Negative Sentiment



Top N-grams in Positive Sentiment



Top N-grams in Neutral Sentiment



1. General Trends (All Tweets)

- Conversations largely center on product launches, pop-up events, and tech unveilings, particularly Apple's retail experiences and Google's social platform rollouts.
- Frequent unigrams such as *ipad*, *apple*, and *google* confirm brand-dominated discourse.
- Multi-word terms highlight Apple's physical brand experiences (e.g., *apple store*, *apple pop store*) and Google's digital innovations (e.g., *new social network*, *google launch*).

major).

- In essence, Apple leads in experiential and retail-driven mentions, while Google dominates discussions around digital innovation and connectivity.

2. Brand-Specific Trends

Apple

- Top trigrams like *apple pop store* and *store downtown austin* emphasize experiential marketing and event-driven buzz.
- Frequent terms such as *ipad* and *iphone app* point to strong user engagement with Apple's product ecosystem.
- Overall, Apple conversations are anchored in tangible experiences and design-centered discussions, often amplified by store events and product showcases.

Google

- Common mentions like *social network*, *circle*, and *launch major new* signal anticipation around Google+ and emerging digital tools.
- Co-occurrence of *marissa mayer* and *google circle* reflects executive-driven product publicity and leadership influence on brand perception.
- Google's discourse appears more conceptual and innovation-oriented, focusing on future potential rather than direct user experience.

3. Sentiment-Specific Trends

Positive Sentiment

- Dominated by launch excitement and product admiration, particularly for Apple (*apple store*, *pop store*, *ipad launch link*).
- Google's positive mentions such as *google map* and *new social network* show optimism and curiosity toward new digital experiences.
- Pop-up stores and launch trigrams underscore enthusiasm, novelty, and brand vitality.

Negative Sentiment

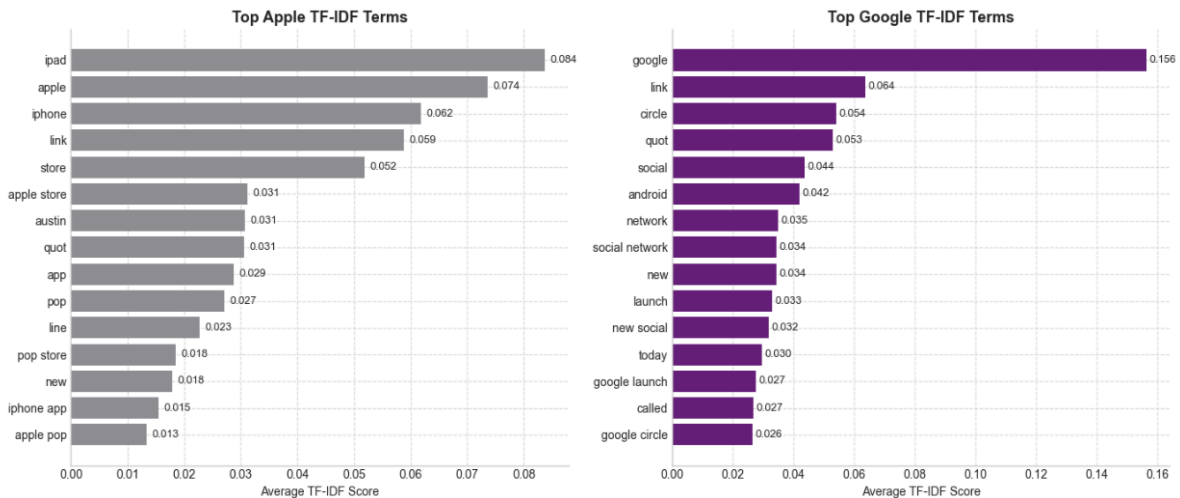
- Negative terms like *design headache*, *battery*, and *company america* point to user frustrations, especially with Apple's products.
- Phrases such as *fascist company america* and *novelty apps fade fast* reveal criticism of over-marketing and brand fatigue.
- While Google faces less negativity overall, its critiques often involve skepticism toward new ventures like Google Circle.

Neutral Sentiment

- Neutral tweets tend to be factual and news-style, reporting events or product details without emotional tone.

- Common phrases like *ipad design headache* and *social network called* indicate objective coverage of product updates or technical issues, rather than opinion-driven engagement.

TF-IDF Analysis



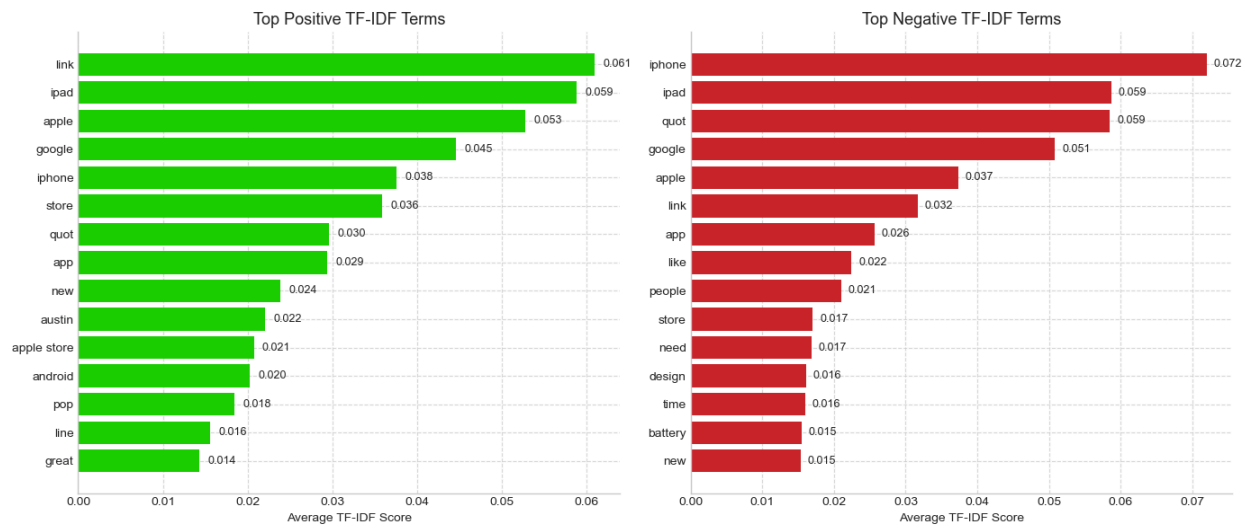
Apple

- Conversations around Apple are driven by terms like *ipad*, *iphone*, *store*, and *app*, underscoring a strong product-centric and event-driven excitement. Mentions of *apple store*, *pop store*, and *austin* highlight retail buzz and localized launch enthusiasm. The overall tone is experiential and lifestyle-oriented. People are talking about what they're buying, seeing, or trying.

Google

- For Google, dominant terms such as *google*, *android*, *circle*, *network*, and *map* focus on connectivity, innovation, and digital integration. Phrases like *social network*, *google map*, and *launch* point to anticipation around new features and platforms. The tone is more functional and tech-oriented, contrasting Apple's more experience-driven narrative.

Most common terms for positive and negative sentiments



Positive Sentiment

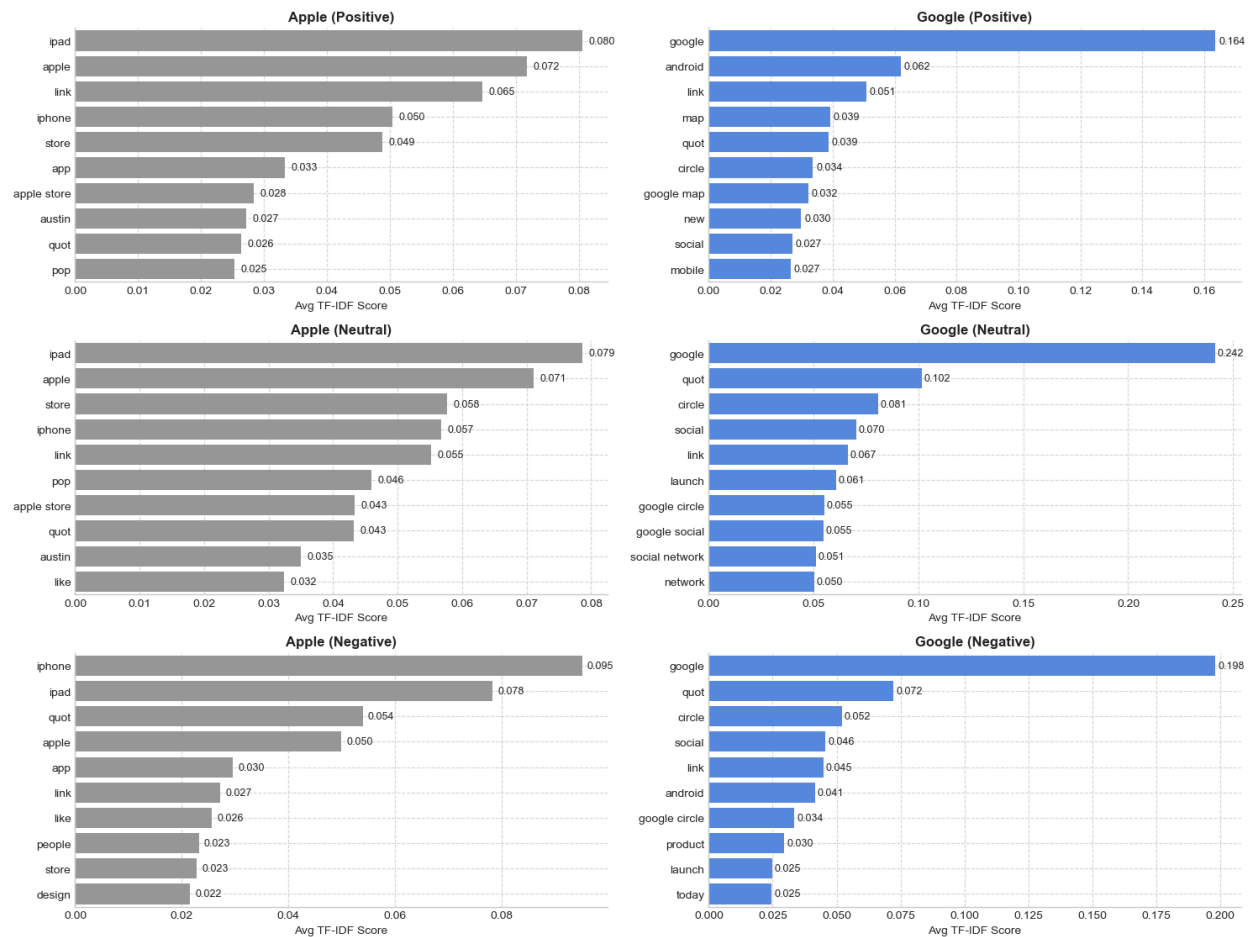
- Frequent positive terms such as *ipad*, *apple*, *google*, *iphone*, and *store* convey **enthusiasm around new product launches and user experiences**. Words like *great*, *new*, and *austin* add a layer of **excitement and satisfaction**, particularly highlighting **Apple's ecosystem appeal** and **Google's innovation-driven interest**.

Negative Sentiment

- Negative chatter revolves around *iphone*, *ipad*, *google*, and *battery*, often paired with *design*, *need*, and *time*. This points to **user frustration with performance issues, design concerns, or release fatigue**, suggesting that criticism stems from **active users rather than disengaged audiences**.

Top sentiment terms associated with each brand

Top TF-IDF Terms by Brand and Sentiment



Apple

- Conversations around Apple are anchored in **tangible products**, with *ipad*, *iphone*, *store*, and *app* dominating across all sentiment categories. This reflects a **deeply product-centric dialogue**, where even criticism revolves around the same flagship devices, signifying **strong engagement paired with high consumer expectations**.

Google

- Discussions about Google gravitate toward **services and ecosystems**, featuring terms like *google*, *android*, *map*, *circle*, and *social network*. The consistent dominance of the term *google* across sentiments underscores the brand's **broad identity strength and ecosystem influence**, framing it as a **platform rather than just a product** in public perception.

Modeling

Target Definition & Preprocessing

During preprocessing, the “sentiment” and “brand” classes were encoded to make them numeric, after which they were added to the TF-IDF matrix for modeling.

Our target variable was the sentiment. It was separated from all other features.

Approach

Several algorithms were trained and tuned:

- Baseline model: Logistic Regression and Decision Trees
- Ensemble models (main focus): Random Forest and XGBoost

Three strategies were applied:

1. Vanilla training.
2. Dimensionality Reduction using Singular Value Decomposition (SVD)
3. Class weighting to counter class imbalance.
4. SMOTE (Synthetic Minority Oversampling Technique) to resample the training data.

For ensemble models, hyperparameter optimization was performed using GridSearchCV with 5-fold cross-validation.

Evaluation Metrics

The models were evaluated using:

- Test accuracy (overall correctness).
- Recall for the minority classes (neutral and negative sentiments).
- Precision and F1-score (balancing false positives and false negatives).
- Confusion matrices for intuitive error breakdown.
- ROC-AUC Curve.

Results (Tuned Models)

Metric	XGBoost (Tuned)	Random Forest (Tuned)
Training Accuracy	98.21%	95.25%

Testing Accuracy	84.47%	78.08%
Macro F1-Score	0.46	0.46
Macro ROC-AUC	0.768	0.705

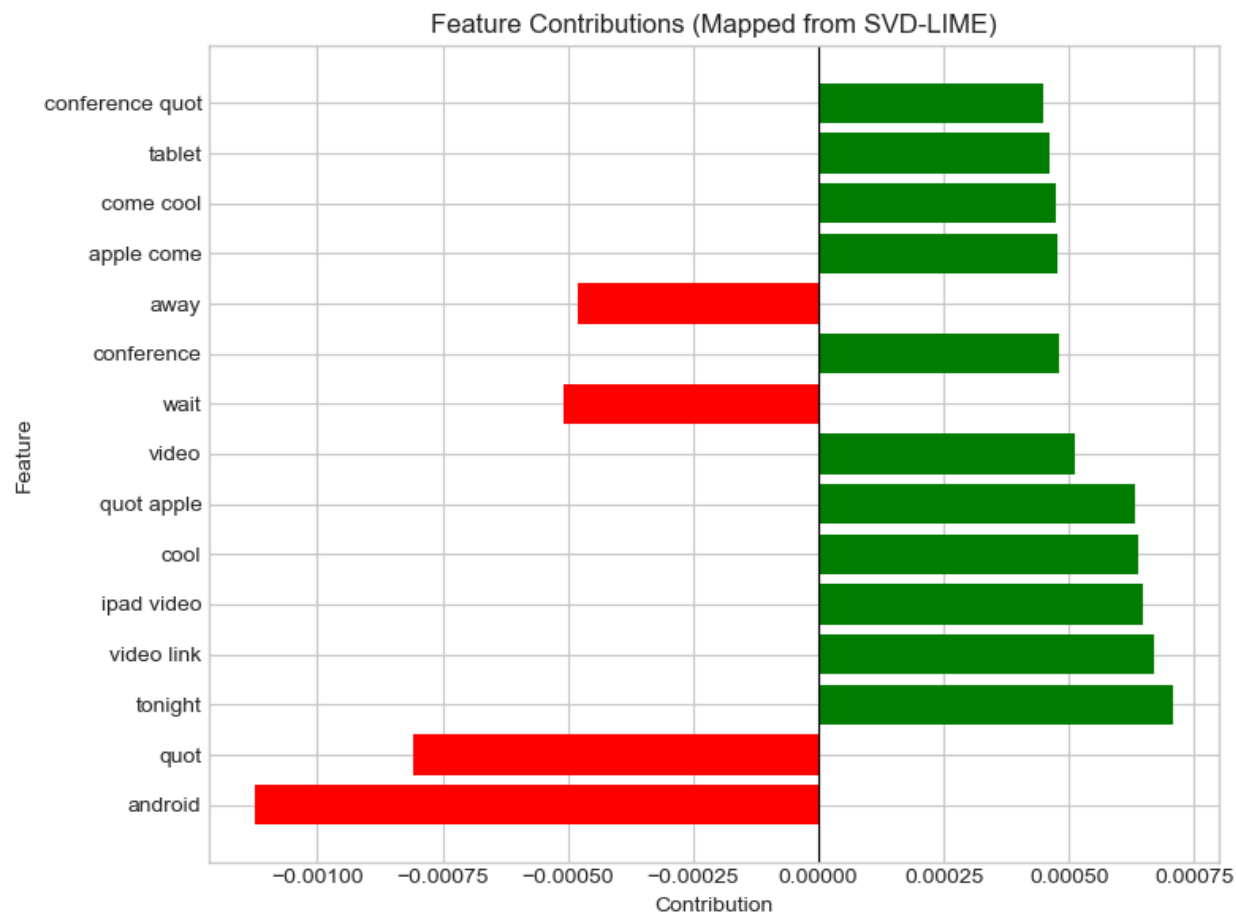
Insights

- **XGBoost** demonstrated **superior generalization**, maintaining high test accuracy and the best macro ROC-AUC score.
- It shows **strong predictive stability**, especially for the dominant class (**functional**), while still outperforming Random Forest in minority class recall.
- **Random Forest**, though solid, displayed **slightly lower test performance** and weaker discrimination across all classes.

Model Behavior Summary

- **XGBoost** captured **non-linear decision boundaries** effectively, leveraging ensemble depth and boosting iterations to minimize error.
- The model achieved **98.2% training accuracy**, indicating a well-fitted but not overly memorized pattern.
- **Random Forest**, while robust and interpretable, showed **slight overfitting** and poorer generalization under class imbalance.

Feature Importance and Interpretation



We analyzed 595 of our 1,001 instances using our XGBoost model and applied **LIME** to understand the contribution of individual features to the model's predictions. Since the model was trained on **SVD-transformed features**, contributions were mapped back to the original features for interpretability.

Key Insights

Top contributing features (positive contributions increase predicted probability; negative contributions decrease it):

Feature	Contribution Direction	Description
apps	Positive	Slightly increases likelihood of the predicted outcome

android	Positive	Slightly increases likelihood
demo	Positive	Slightly increases likelihood
need	Positive	Slightly increases likelihood
great	Negative	Slightly decreases likelihood
app	Negative	Slightly decreases likelihood
new iphone	Negative	Slightly decreases likelihood
line ipad	Negative	Slightly decreases likelihood
playing	Negative	Slightly decreases likelihood
app link	Negative	Slightly decreases likelihood

Interpretation:

The prediction is influenced by a mix of **positive and negative signals**. Features such as *apps*, *android*, and *demo* push predictions toward the predicted class, while features like *app*, *new iphone*, *line ipad*, and *playing* pull predictions away.

Takeaways:

- The model is particularly sensitive to **app- and product-related terms**, as well as general engagement words.
- These insights can inform **feature selection, product recommendations, and content optimization**, while also enhancing transparency into the model's decision-making process.

Recommendations

Given its **higher testing accuracy, superior ROC–AUC (0.768)**, and **more balanced class discrimination**, **XGBoost** is the recommended model for deployment.

It provides the best trade-off between predictive power, generalization, and operational efficiency, making it the most reliable choice for production use.

Model Limitations

While the tuned XGBoost model achieved strong overall performance; particularly in recall and ROC-AUC, several limitations remain that constrain its full generalizability and interpretability:

1. Class Imbalance Sensitivity

- Despite the use of class weights, minority classes (especially class 1- neutral) remain underrepresented in model predictions, leading to lower recall and precision for these cases. This suggests that further balancing or targeted sampling may be required.

2. Feature Representation Constraints

- The use of SVD for dimensionality reduction, while effective in reducing computational cost, may have caused the loss of subtle yet discriminative information. This could limit the model's ability to capture nuanced patterns in the original feature space.

3. Overfitting Risk

- The gap between training accuracy (98.2%) and test accuracy (84.5%) indicates mild overfitting. The model has likely learned noise or dataset-specific characteristics that do not generalize well to unseen data.

4. Limited Temporal Awareness

- The model does not account for temporal or contextual evolution (e.g. changes in trends, seasonality, or time-dependent shifts). Predictions assume feature distributions remain static over time.

5. Interpretability Challenge

- While XGBoost offers feature importance metrics, the underlying decision boundaries remain complex and opaque compared to simpler models like logistic regression, making it harder to explain predictions to non-technical stakeholders.

6. Dependence on Preprocessing Consistency

- The model's performance heavily depends on consistent preprocessing steps (scaling, dimensionality reduction). Any deviation during deployment could lead to prediction drift or misclassification.

Future Work & Improvement

To enhance model performance, interpretability, and scalability, several steps can be taken in future iterations:

1. Advanced Explainability Techniques:

- Implement SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-Agnostic Explanations) to improve transparency and stakeholder understanding of feature impacts.

2. Model Ensemble or Stacking:

- Combine XGBoost with complementary models like Random Forest or LightGBM using stacking or blending to capture diverse feature interactions and reduce variance.

3. Enhanced Feature Engineering:

- Incorporate interaction terms, domain-specific indicators, and statistical aggregates to improve signal quality and model discrimination power.

4. Robust Cross-Validation:

- Use stratified k-fold cross-validation to ensure balanced class representation across folds, reducing variance in performance metrics.

5. Hyperparameter Optimization at Scale:

- Explore Bayesian optimization (e.g Optuna) for smarter, faster parameter tuning instead of exhaustive grid search.

6. Handling Class Imbalance More Effectively:

- Test advanced resampling strategies such as SMOTEENN or ADASYN, or adjust `scale_pos_weight` dynamically to better handle minority classes.

7. Model Deployment & Monitoring:

- After deployment, continuously monitor model drift and retrain periodically using fresh data to maintain accuracy over time.